

Marianna Rosso, PhD

Biostatistician & Data Scientist | Statistical Computing, Reproducible Research & Evidence Synthesis

Im Langacker 19, 8304 Wallisellen, Switzerland | +41 (0)78 608 5177 | marianna.rosso3@gmail.com

PROFILE

Data scientist and postdoctoral researcher with 7+ years of experience in statistical computing, reproducible research workflows, and biomedical data science. Proven track record designing R and R Shiny tools for evidence synthesis, data visualization, and automated reporting, paired with a record of peer-reviewed publications in statistical and biomedical methodology (PRISMA, ARRIVE). Experienced collaborating with multidisciplinary, international teams and mentoring junior researchers in statistical software and reproducible research practices.

SKILLS

Core Competencies: Project Management, Scientific Leadership, Stakeholder Management, Cross-functional Collaboration, Mentoring & Training, Process Improvement

Data & Statistical Methods: Statistical Computing, Evidence Synthesis & Meta-Analysis, Reproducible Workflows, ETL / Data Pipeline Design, Data Architecture & Modelling, Data Quality Assurance, FAIR Principles

Tools & Languages: R (advanced), Python, BASH, SQL, R Shiny / Quarto, Git/GitHub, Tableau, VS Code

EMPLOYMENT HISTORY

Data Scientist & Project Manager | University Hospital Zurich | *Jun 2023 – May 2026*

- Designed, built, and maintained data pipelines and data architecture for real-world ophthalmology research datasets.
- Developed data analyses, visualisations, R Shiny dashboards, and automated reporting workflows; collaborated with clinicians to interpret real-world clinical data for scientific publications.
- Introduced API-based data integration to streamline data ingestion, processing, and reporting.
- Oversaw data quality assurance and governance across multiple research projects and datasets.
- Implemented Git-based version control and reproducible coding practices across distributed international teams.
- Produced publication-quality figures, tables, and visualisations, and contributed to scientific manuscripts and conference presentations.

Postdoctoral Researcher | Epidemiology, Biostatistics, and Prevention Institute, University of Zurich | *Nov 2022 – Jan 2026*

- Developed automated data extraction and analysis pipelines, including R Shiny applications and dashboards, to accelerate evidence synthesis across large biomedical and preclinical datasets.
- Authored peer-reviewed publications, systematic reviews, and meta-analyses in biomedical research, applying rigorous methodological standards (PRISMA, ARRIVE).
- Interpreted complex clinical and preclinical data, generating evidence to support methodological improvements in biomedical science.
- Supervised and mentored junior researchers in statistical software development, scientific writing, and reproducible research practices.
- Secured external funding and managed cross-institutional projects and training initiatives.

Co-founder & Data Consultant | SheViz, Self-employed | *Mar 2021- Jan 2025*

- Advised academic and non-academic clients on data pipeline design, statistical analysis, data visualisation, and FAIR data implementation.
- Designed and delivered training on open science workflows, data management strategy, and research integrity.

PhD Researcher | Veterinary Public Health Institute, University of Bern | *Dec 2018 – Jun 2022*

- Conducted large-scale systematic reviews and meta-analyses, extracting and synthesising data across hundreds of studies, applying risk-of-bias and statistical reliability assessments.
- Taught and mentored Master's students in research methods, statistics, and scientific writing.

SELECTED PROJECTS

- Meta-Analyze — Interactive R Shiny dashboard demonstrating a live meta-analysis workflow for evidence-synthesis methods. ([link](#))
- CLEVER-MS — R Shiny tool for extraction and validation of clinical trial data in multiple sclerosis research. ([link](#))
- Mouse Anxiety Behavioural Tests Dashboard — R Shiny dashboard built from a systematic review and meta-analysis on the validity of mouse behavioural tests. ([link](#))
- Designed and deployed several website with Quarto, using CSS and Java language. (Example here: [link](#))
- Led international scientific programmes with end-to-end responsibility for stakeholder management, planning, and delivery; secured CHF 46,460+ and USD 5,000 in external funding.

SELECTED PUBLICATIONS (full list here: [link](#))

- Menon JLM - **Rosso M**, et al. (2025): The efficacy of candidate HIV vaccines in non-human primates and humans: a systematic review. In medRxiv
- **Rosso M**, et al. (2024). Evidence of HARKing in mouse behavioural tests of anxiety. Royal Society Open Science.
- Ineichen BV, **Rosso M**, Macleod MR (2023). From data deluge to publomics: How AI can transform animal research. Lab Animal.
- **Rosso M**, Doneva S, Howells DW, Leenaars CHC, Ineichen BV (2023): Summer school for systematic reviews of animal studies: fostering evidence-based and rigorous animal research. In ALTEX – Alternatives to animal experimentation 41 (1)
- **Rosso M**, et al. (2022). Reliability of Common Mouse Behavioural Tests of Anxiety: Systematic Review and Meta-Analysis on the Effects of Anxiolytics. Neuroscience and Biobehavioural Reviews.

SELECTED OUTREACH

- Lecturer: Graphical Statistics course, GCB Graduate School, University of Bern
- Award: YOU-WC11 Workshop for best presentations from Early Career Researchers.
- Talk: 11th edition of the World Congress on Alternatives and Animal Use in the Life Sciences – Sensitivity of mouse behavioural tests of anxiety to anxiolytic drugs approved for treatment of anxiety in humans: a systematic review and meta-analysis
- Poster presentation: 3Rs Day 2023 – Launching CAMARADES Switzerland – Systematic Reviews improve translatability of biomedical findings

EDUCATION

PhD Degree, University of Bern, Switzerland

Biomedical Sciences, Dec 2018 – Jun 2022

Master's Degree, University of Bern, Switzerland

Ecology and Evolution, specialization in Evolutionary Behaviour, Feb 2016 – Sep 2018

Bachelor's Degree, Università degli Studi di Torino, Italy

Biology, specialization in Ecology, Oct 2011 – Oct 2015